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Abstract. one day i'll write an abstract

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1. Introduction

Intro and motivation

2. The Model

We consider a static duopoly game of incomplete information. Two symmetric firms, indexed by $i \in \{1, 2\}$, compete by simultaneously choosing a level of strategic deception $d_i \in [0, \bar{d}]$, where $\bar{d}$ represents the structural upper bound of feasible deceptive practices.

2.1. Consumer Behavior and Demand Microfoundations

The market consists of a continuum of heterogeneous consumers normalized to a total mass of one. 'Deception' in this market is defined as an action taken by a firm that increases the perceived attractiveness of a product to consumers without intrinsically changing the product itself. Each consumer is characterized by a private tolerance threshold τ , which represents the maximum level of deception they will tolerate before completely rejecting a firm's product. We assume that consumer tolerance is uniformly distributed across the population up to an aggregate market limit θ , such that $\tau \sim U[0, \theta]$.

A consumer facing a firm playing deception level d_i evaluates their options according to their decision utility:

Assumption 1 (Consumer Decision Utility). The utility a consumer with private tolerance τ derives from purchasing from firm i is given by

$$\tilde{U}(d_i, \tau) = \begin{cases} v(d_i) & \text{if } d_i \leq \tau \\ 0 & \text{if } d_i > \tau \end{cases} \quad (1)$$

with $v'(d_i) > 0$. Since we only model consumer choice, experienced utility is not modeled directly. However, we assume that it is a decreasing function in deception, i.e. deception inflates the perceived value of the product when making the choice to buy, but reduces actual experienced utility once the consumer eventually realizes they've been deceived.

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A higher level of d_i artificially inflates the product's perceived attractiveness, driving up the consumer's perceived utility. This psychological illusion persists continuously until the firm overreaches and crosses the consumer's private tolerance threshold τ . At this point, the consumer detects the deception and ceases to buy from the firm.

Let $d_H = \max\{d_1, d_2\}$ and $d_L = \min\{d_1, d_2\}$ denote the high and low deception firms. Given that consumers maximize utility subject to their tolerance constraint, the aggregate demand schedules q_H and q_L are determined by the realization of θ .

$$q_H(d_H; \theta) = \begin{cases} \frac{\theta - d_H}{\theta}, & d_H \leq \theta, \\ 0, & d_H > \theta, \end{cases} \quad q_L(d_L, d_H; \theta) = \begin{cases} \frac{d_H - d_L}{\theta}, & d_H \leq \theta, \\ \frac{\theta - d_L}{\theta}, & d_L \leq \theta < d_H, \\ 0, & d_L > \theta. \end{cases} \quad (2)$$

2.2. Firm Payoffs and the Optimization Problem

Firms are risk-neutral and maximize expected profits. Deception operates as a double-edged sword: it reduces consumer demand, but increases the revenue extracted per retained customer. We formalize this revenue a firm can extract *because* deception inflates perceived value as $r(d_i) = d_i^\gamma$, where $\gamma > 0$. We will make two useful assumptions on r :

1. $r(0) = 0$, and r is positive, differentiable and strictly increasing on $(0, \bar{d}]$,
2. $\frac{r(d)^2}{r'(d)}$ is strictly increasing on $(0, \bar{d}]$.

The profit function for firm i is given by:

$$\pi_i(d_i, d_j; \theta) = r(d_i) \cdot q_i(d_i, d_j; \theta) \quad (3)$$

3. Solving the model with complete information

3.1. Monopoly benchmark

A single firm in this environment would maximize $\pi = d^\gamma \frac{\theta - d}{\theta}$. Maximizing this, we get the result

$$d^{mon} = \frac{\gamma}{\gamma + 1} \theta, \quad \pi^{mon} = \frac{\gamma^\gamma}{(\gamma + 1)^{\gamma+1}} \theta^\gamma$$

3.2. Pure Strategy Nash Equilibrium

Proposition 1. *For any strictly increasing continuous $r(d)$, the game has no pure strategy Nash equilibrium.*

Proof. If $d_1 = d_2 = d$ and $d < \theta$, both firms share the consumers tolerant to d equally, each earning $\pi_i = r(d) \frac{\theta-d}{2\theta}$. Consider a deviation to $d + \varepsilon$. Profit for the deviator now becomes $\pi_i = r(d + \varepsilon) \frac{\theta-d-\varepsilon}{\theta}$ as they no longer have to share the market. As $\varepsilon \rightarrow 0$, $\pi_i = r(d) \frac{\theta-d}{\theta}$, which is twice as high. Therefore, a small positive deviation is strictly profitable.

If $d_L < d_H$ and $d_H > \theta$, the high firm earns zero and can deviate to below θ to earn positive profits. If $d_H < \theta$, we know that at equilibrium, $\pi_H(d_H) > \pi_H(d) \forall d$. In particular, $\pi_H(d_H) > \pi_H(d_L)$ for the high firm. Now for the low firm, it earns $\pi_L = r(d_L) \frac{d_H-d_L}{\theta}$. But notice this is strictly less than the numerator of $\pi_H(d_L) = r(d_L) \frac{\theta-d_L}{\theta}$ as $d_H < \theta$. Therefore, we get $\pi_H(d_H) > \pi_H(d_L) > \pi_L(d_L)$. If the low firm plays $d_H + \varepsilon$, it becomes the new high firm and earns a higher profit than being the low firm, giving it an incentive to deviate. $\square$

4. Mixed Strategy Nash Equilibrium

4.1. Indifference condition

Suppose the firm earns the same π^* across the entire support $[\underline{s}, \bar{s}]$. If $F(d)$ denotes the CDF from which the firms draw actions, then their expected profit becomes

$$\mathbb{E}(\pi) = r(d) \left[F(d) \frac{\theta-d}{\theta} + (1-F(d)) \mathbb{E} \left[\frac{x-d}{\theta} | x > d \right] \right] = \pi^* \quad \forall d \in [\underline{s}, \bar{s}]$$

To solve for MSNE, we need to set $\mathbb{E}'(\pi) = 0$. We'll do so by differentiating elements of $\mathbb{E}(\pi)$ one by one. If $f(d)$ is the density of $F(d)$, we can rewrite the conditional expectation. Let's call it the expected residual demand, represented by $R(d)$.

$$(1-F(d)) \mathbb{E} \left[\frac{x-d}{\theta} | x > d \right] = \int_d^{\bar{s}} \frac{x-d}{\theta} f(x) dx \equiv R(d)$$

4.2. Solving for the density

Differentiating it, we get

$$\begin{aligned} \frac{d}{dd} \int_d^{\bar{s}} \frac{x-d}{\theta} f(x) dx &= \frac{1}{\theta} \left[-(d-d)f(d) + \int_d^{\bar{s}} \frac{\partial}{\partial d} [(x-d)f(x)] dx \right] \\ &= \frac{1}{\theta} \left[\int_d^{\bar{s}} -(f(x)) dx \right] \\ &= \frac{1}{\theta} [-(F(\bar{s}) - F(d))] \\ \implies R'(d) &= \frac{-(1-F(d))}{\theta} \end{aligned}$$

Next, differentiating $F(d) \frac{\theta-d}{\theta} + R(d)$ gives us

$$\begin{aligned} \frac{d}{dd} \left[F(d) \frac{\theta-d}{\theta} + R(d) \right] &= f(d) \frac{\theta-d}{\theta} - \frac{F(d)}{\theta} + R'(d) \\ &= f(d) \frac{\theta-d}{\theta} - \frac{F(d)}{\theta} - \frac{1-F(d)}{\theta} \\ &= \frac{f(d)(\theta-d)}{\theta} - \frac{1}{\theta} \end{aligned} \tag{4}$$

Finally, differentiating $\mathbb{E}(\pi)$ yields

$$\mathbb{E}'(\pi) = r'(d) \left[F(d) \frac{\theta - d}{\theta} + R(d) \right] + r(d) \left[\frac{f(d)(\theta - d) - 1}{\theta} \right] = 0$$

Since $\left[F(d) \frac{\theta - d}{\theta} + R(d) \right] = \frac{\pi^*}{r(d)}$, we can clean up this expression to get

$$\begin{aligned} \mathbb{E}'(\pi) &= \frac{r'(d)\pi^*}{r(d)} + \frac{r(d)}{\theta} [f(d)(\theta - d) - 1] = 0 \\ \implies \frac{r'(d)\pi^*\theta}{r(d)^2} + f(d)(\theta - d) - 1 &= 0 \end{aligned}$$

Therefore, we get the density

$$f(d) = \frac{1}{\theta - d} \left[1 - \frac{r'(d)\pi^*\theta}{r(d)^2} \right] \quad (5)$$

4.3. Identifying the optimal profit

Our next goal will be to pin down π^* . To do so, we can evaluate it at $F(\underline{s})$ and $F(\bar{s})$, values of F we know. At $d = \underline{s}$, $F(\underline{s}) = 0$, giving us

$$\begin{aligned} \pi^* &= r(\underline{s}) \int_{\underline{s}}^{\bar{s}} \frac{x - \underline{s}}{\theta} f(x) dx \\ &= \frac{r(\underline{s})}{\theta} \left[\int_{\underline{s}}^{\bar{s}} x f(x) dx - \underline{s} \int_{\underline{s}}^{\bar{s}} f(x) dx \right] \\ &= \frac{r(\underline{s}) [\mathbb{E}[d] - \underline{s}]}{\theta} \end{aligned}$$

For any $d < \underline{s}$, we know that $F(d) = 0$, so we get a payoff of the same form $\pi(d) = \frac{r(d) [\mathbb{E}[d] - d]}{\theta}$. Since this d is below the support, it is not in the optimal interval for the MSNE, and thus it must earn a strictly lower profit than $\pi(\underline{s}) = \pi^*$. In other words, $\underline{s}$ maximizes profits, and thus also maximizes $\frac{r(d) [\mathbb{E}[d] - d]}{\theta}$ for $d \leq \underline{s}$. Differentiating, the FOC is $r'(\underline{s}) [\mathbb{E}(d) - \underline{s}] - r(\underline{s}) \geq 0$ at the right side boundary, and thus we get

$$\mathbb{E}[d] - \underline{s} \geq \frac{r(\underline{s})}{r'(\underline{s})}$$

Substituting into the expression for π^* , we get

$$\pi^* \geq \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$$

But from the density (Equation 5), $f(\underline{s}) \geq 0 \implies \frac{r^2(\underline{s})}{r'(\underline{s})\theta} \geq \pi^*$. Combining the two inequalities, we get

$$\pi^* = \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$$

At $d = F(\bar{s}) = 1$, $\pi^* = r(\bar{s})\frac{\theta - \bar{s}}{\theta}$, so

$$\pi^* = \frac{r(\bar{s})(\theta - \bar{s})}{\theta}$$

Since $\pi^* > 0$, this establishes that the upper end of the support $\bar{s} < \theta$, so firms never deceive past the market boundary in a complete information MSNE.

Proposition 2 (Expected value of deception). *Under complete information, $\mathbb{E}[d] = \frac{\gamma+1}{\gamma}\underline{s}$.*

Proof. We know $\pi^* = \frac{r(\underline{s})[\mathbb{E}[d] - \underline{s}]}{\theta} = \frac{r^2(\underline{s})}{r'(\underline{s})\theta}$. This gives us $\mathbb{E}[d] = \underline{s} + \frac{r(\underline{s})}{r'(\underline{s})} = \frac{\gamma+1}{\gamma}\underline{s}$. $\square$

4.4. With constant elasticity revenue

To move further, let's use our assumed functional form $r(d) = d^\gamma$. This gives us

$$\pi^* = \frac{\underline{s}^{\gamma+1}}{\gamma\theta} = \frac{\bar{s}^\gamma(\theta - \bar{s})}{\theta}$$

and our density is

$$f(d) = \frac{1}{\theta - d} \left[1 - \left(\frac{\underline{s}}{d} \right)^{\gamma+1} \right]$$

Integrating over the whole support gives us

$$\int_{\underline{s}}^{\bar{s}} \frac{1}{\theta - d} \left[1 - \left(\frac{\underline{s}}{d} \right)^{\gamma+1} \right] dd = 1$$

For simplicity, let $\gamma = 1$. Then, $f(d) = \frac{1}{\theta - d} - \frac{\underline{s}^2}{d^2(\theta - d)}$. Integrating this gives us

$$F(d) = \left(\left(\frac{\underline{s}}{\theta} \right)^2 - 1 \right) \ln(\theta - d) - \left(\frac{\underline{s}}{\theta} \right)^2 \ln(d) + \left(\frac{\underline{s}}{\theta} \right)^2 \frac{\theta}{d} + C$$

Since we know $F(\bar{s}) - F(\underline{s}) = 1$, subtracting gives us

$$(\rho^2 - 1) \ln \left(\frac{\theta - \bar{s}}{\theta - \underline{s}} \right) - \rho^2 \ln \left(\frac{\bar{s}}{\underline{s}} \right) + \rho^2 \theta \left(\frac{1}{\bar{s}} - \frac{1}{\underline{s}} \right) = 1$$

where $\rho \equiv \frac{\underline{s}}{\theta}$. From π^* , we get $\underline{s}^{\gamma+1} = \gamma \bar{s}^\gamma (\theta - \bar{s})$. Therefore, the above expression is an equation in one variable which can be solved numerically. Doing so for $\theta = 1$, we get $\bar{s} = 0.8557$ and thus $\underline{s} = 0.3514$.

Proposition 3 (Scale invariance). *Fix $\gamma > 0$ and let $\theta > 0$ be arbitrary. If F_1 is an equilibrium distribution for a market with tolerance ceiling 1, then the distribution F_θ defined by $F_\theta(d) = F_1(d/\theta)$ is an equilibrium distribution for a market with ceiling θ , and conversely. Consequently*

$$\underline{s}(\theta) = \theta \underline{s}(1), \quad \bar{s}(\theta) = \theta \bar{s}(1), \quad \mathbb{E}_\theta[d] = \theta \mathbb{E}_1[d], \quad \pi^*(\theta) = \theta^\gamma \pi^*(1).$$

Proof. Write actions as $d_i = \theta u_i$ with $u_i \in [0, 1]$. Since the demand schedules depend on deception only through its ratio to the ceiling,

$$q_H = \frac{\theta - d_H}{\theta} = 1 - u_H, \quad q_L = \frac{d_H - d_L}{\theta} = u_H - u_L,$$

so demand in the market with ceiling θ evaluated at $(\theta u_1, \theta u_2)$ equals demand in the market with ceiling 1 evaluated at (u_1, u_2) . Revenue is homogeneous of degree γ , since $r(\theta u) = (\theta u)^\gamma = \theta^\gamma r(u)$. Therefore

$$\pi_i(\theta u_i, \theta u_j; \theta) = r(\theta u_i) q_i(u_i, u_j) = \theta^\gamma \pi_i(u_i, u_j; 1).$$

The map $d \mapsto d/\theta$ is thus a bijection between the action spaces under which every payoff in the market with ceiling θ equals θ^γ times the corresponding payoff in the market with ceiling 1. $\square$

Corollary 1 (Competitive escalation). *Let $\gamma = 1$. In the mixed strategy equilibrium, for every tolerance ceiling θ ,*

$$\underline{s} = 0.3514 \theta, \quad \mathbb{E}[d] = 0.7029 \theta, \quad \bar{s} = 0.8557 \theta,$$

and, relative to the monopoly benchmark $d^{\text{mon}} = \gamma\theta/(\gamma + 1)$,

$$\frac{\mathbb{E}[d]}{d^{\text{mon}}} = 1.406, \quad \frac{\bar{s}}{d^{\text{mon}}} = 1.711, \quad \frac{\pi^*}{\Pi^{\text{mon}}} = 0.494.$$

These ratios are independent of θ . Average deception under duopoly therefore exceeds the monopoly level by roughly 41 per cent, and the highest level played exceeds it by roughly 71 per cent, while equilibrium profit falls to under half the monopoly benchmark.

Proof. The solution of the normalisation condition at $\theta = 1$ gives $\bar{s} = 0.8557$, and $\underline{s} = \sqrt{\bar{s}(1 - \bar{s})} = 0.3514$; the mean follows from $\mathbb{E}[d] = (\gamma + 1)\underline{s}/\gamma$. Rescaling by Proposition 3 gives the stated levels for arbitrary θ . $\square$

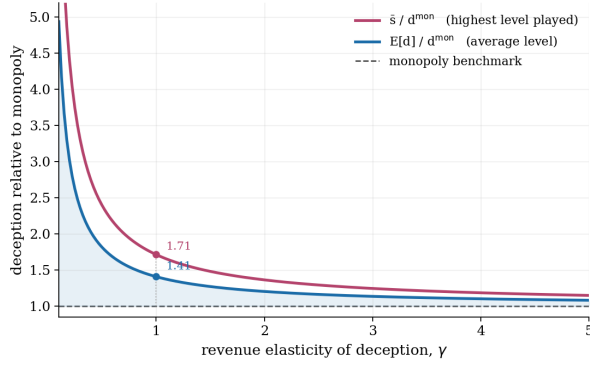
As we sweep across various values of γ , the revenue elasticity of deception, we get the following results:

Figure 1a plots equilibrium deception relative to the monopoly benchmark, and Figure 1b the support of the equilibrium distribution, both as fractions of θ .

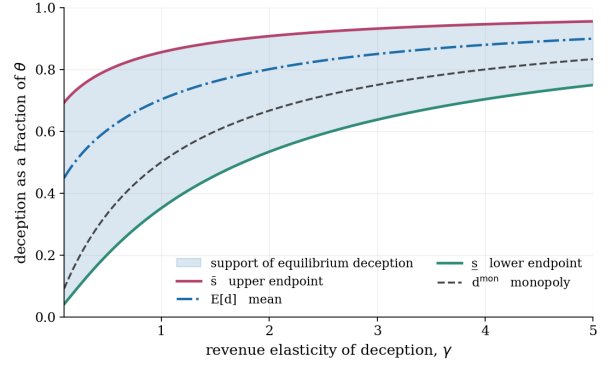
The ratio wrt monopoly falls in γ and diverges as $\gamma \rightarrow 0$. From Equation 4, the derivative of expected demand is $D'(d) = \frac{f(d)(\theta - d)}{\theta} - \frac{1}{\theta}$. It tells us that raising d forfeits the mass $1/\theta$ of consumers whose tolerance is just crossed, but overtakes any rival and becoming the high firm with probability $f(d)$. A monopolist faces the first effect and not the second, only deceiving for the added revenue from γ . As $\gamma \rightarrow 0$ the monopolist therefore stops deceiving, $d^{\text{mon}} \rightarrow 0$, while duopolists retain the incentive to leapfrog. Even though deception is increasing in γ , as the duopolists still retain some incentive to deceive while monopolists do not, the ratio $\frac{\mathbb{E}[d]}{d^{\text{mon}}}$ blows up as $\gamma \rightarrow 0$.

Table 1: Equilibrium deception across revenue elasticities

γ	Equilibrium distribution				Relative to monopoly		
	$\underline{s}$	$\mathbb{E}[d]$	$\bar{s}$	$\bar{s} - \underline{s}$	$\frac{\mathbb{E}[d]}{d^{\text{mon}}}$	$\frac{\bar{s}}{d^{\text{mon}}}$	$\frac{\pi^*}{\Pi^{\text{mon}}}$
0.25	0.104	0.522	0.744	0.640	2.610	3.722	0.444
0.50	0.201	0.604	0.798	0.596	1.812	2.393	0.470
0.75	0.283	0.661	0.832	0.549	1.542	1.941	0.484
1.00	0.351	0.703	0.856	0.504	1.406	1.711	0.494
1.50	0.457	0.762	0.887	0.431	1.269	1.479	0.506
2.00	0.534	0.801	0.908	0.374	1.201	1.361	0.513
3.00	0.638	0.850	0.932	0.295	1.133	1.243	0.522
4.00	0.704	0.880	0.946	0.242	1.100	1.183	0.527
5.00	0.750	0.900	0.955	0.206	1.079	1.147	0.530



(a) Deception relative to the monopoly benchmark



(b) The support of equilibrium deception

Figure 1: Comparative statics in the revenue elasticity γ .

On the other hand, the support narrows as γ rises. This is because across the support, we must earn the same π^* . In our model, this works because the gain from increased deception perfectly cancels out the loss from lost market share on $[\underline{s}, \bar{s}]$. But as γ rises, revenue extracted increases exponentially. Thus, as γ rises, the revenue gains are so significant there may not be enough market share to compensate the lower end of the support, making it narrower.